

Carl Matthew T. Arzadon

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Skills

Languages: JavaScript, Python, HTML, CSS, C, Java

Frameworks and Libraries: Laravel, Flask, Tailwind CSS, Bootstrap, PyQt5, SQL Alchemy

Databases: SQLite, MySQL, PostgreSQL

Embedded and Hardware: Arduino Uno, ESP8266, ESP32, Soldering

Software and Tools: Git, Github, KiCad

Experience

Student Intern, Web Developer

June 2025 – August 2025

Department of the Interior and Local Government - CAR

Baguio, Benguet

- Developed and deployed a Recruitment, Selection, and Placement (RSP) web app using Laravel, Tailwind CSS, and MySQL.
- Designed the system database through ER modeling and implemented a normalized schema using Laravel Eloquent ORM.
- Built a real-time exam monitoring system with examinee status tracking, administrative controls, and tab-switch detection using Laravel Livewire and Reverb.
- Participated in stakeholder demonstrations, communicating system functionality to technical and non-technical audiences and incorporating feedback into development.

Projects

Grading System | Python, PyQt5, SQLite, SQL Alchemy

github.com/homusys/grading-system

- Led the database design phase by designing the entity relationship diagrams and implementing a normalized relational database using SQLite and SQL Alchemy ORM.
- Worked closely with team members on integrating the backend data operations with the desktop application's GUI and assisted in debugging and resolving issues across multiple application modules.
- Developed background worker threads for database transactions in order to prevent the application from freezing during long-running operations.

Wireless Sensor-Based Traffic Control System | C/C++ , Arduino Framework

github.com/homusys/wireless-sensor-based-traffic-control-system

- Collaborated with stakeholders to define the functional requirements and allocate responsibilities across multiple Arduino micro-controllers.
- Developed a distributed traffic control system featuring ultrasonic sensor-based vehicle and pedestrian detection, sensor-driven control logic, and NRF24L01 wireless communication between controllers.

QR Code Attendance Monitoring System | Python, PyQt5, OpenCV, SQLite, SQL Alchemy, ESP32-CAM

github.com/homusys/qr-attendance

- Built an attendance monitoring system using PyQt5 and OpenCV, integrating an ESP32-CAM via socket-based communication with an SQLite database for attendance tracking.

Certificates

- Javascript Essentials 1 and 2 - Cisco Networking Academy
- Python Essentials 1 - Cisco Networking Academy
- Mechatronics Servicing NCII - TESDA: Technical Education and Skills Development Authority
- Civil Service Examination – Professional Level Passer (Examinee No. 110301)

Education

Pangasinan State University, Urdaneta Campus – BS Computer Engineering

Expected July 2026